Directional evidence for retained main and secondary pathways

	ssGSEA	GSEA direction	Held-out projects	Three trainings	Composition proxy	GSEA FDR	Project direction
Thymus: DNA repair	yes	yes	yes	yes	yes	<0.001	5/5 lower
Thymus: RHOA cytoskeletal cycle	yes	yes	yes	yes	yes	0.154	5/5 lower
Thymus: Lymphoid-stromal interactions	no	no	yes	yes	yes	1.000	5/5 lower
Skin: Chromatin-modifying enzymes	yes	yes	yes	yes	yes	<0.001	3/4 lower
Skin: DNA repair	yes	yes	yes	yes	yes	<0.001	3/4 lower
Skin: Hedgehog signaling	yes	yes	yes	yes	yes	0.280	3/4 lower
Skin: Sphingolipid metabolism	yes	yes	yes	yes	yes	0.897	3/4 lower
Skin: Cell-cell junction organization	no	yes	yes	yes	yes	0.007	3/4 lower
Liver: MHC class II antigen presentation	yes	yes	yes	yes	yes	0.121	8/9 lower
Liver: T-cell receptor signaling	yes	yes	yes	yes	yes	0.051	8/9 lower
Spleen: T-cell receptor signaling	yes	yes	yes	yes	yes	0.042	5/5 lower
Spleen: Neutrophil degranulation program	yes	yes	yes	yes	yes	<0.001	5/5 lower
Spleen: C-type lectin receptor signaling	yes	yes	yes	yes	yes	0.017	5/5 lower
Kidney (secondary): ECM proteoglycans	yes	yes	yes	yes	yes	0.156	5/6 higher
Kidney (secondary): WNT signaling	yes	yes	yes	yes	yes	1.000	6/6 higher
Kidney (secondary): IGF transport and uptake	yes	yes	yes	yes	yes	0.433	5/6 higher

Green cells indicate directional support. GSEA FDR is reported separately because the five-check framework does not use FDR as a binary gate. Kidney directions pass the five directional checks but remain exploratory because GSEA FDR is above 0.05 and composition adjustment attenuates the effects.